

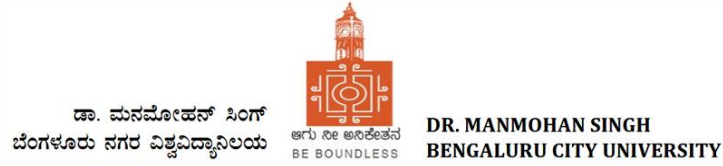
Machine Learning-Based Disease Prediction with Mutation Awareness

A project report submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF COMPUTER APPLICATIONS

of

Dr. Manmohan Singh Bengaluru City University



By
NANDINI P. V.
U18GH23S0113

Under the Guidance of

Mrs. PRATHIBA P. M.
Assistant Professor



SESHADRIPURAM COLLEGE

27, Nagappa Street, Seshadripuram, Bengaluru-560 020 Affiliated to Dr. Manmohan Singh
Bengaluru City University ISO 9001:2015 Certified Recognized Under Section 2(f) and 12(B)
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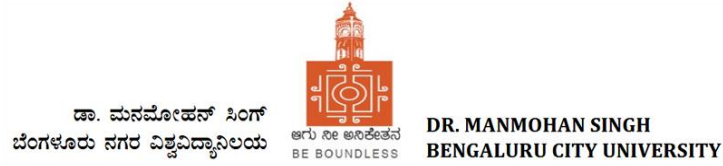
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CERTIFICATE

This is to Certify that **NANDINI P.V.** bearing **U18GH23S0113** has completed her final semester project work entitled “**Machine Learning-Based Disease Prediction with Mutation Awareness**” as a partial fulfillment for the award of Bachelor of Computer Applications degree, during the academic year 2025-26 under my supervision.

Signature of Internal Guide
Mrs. Prathibha P. M.

Signature of HOD
Prof. Rajeshwari V

External Examiner: Internal Examiner:

DECLARATION

I, **NANDINI P. V.**, student of 6th semester BCA, **SESHADRIPURAM COLLEGE**, bearing USN **U18GH23S0113** hereby declare that the project entitled **Machine Learning-Based Disease Prediction with Mutation Awareness** has been carried out by me under the supervision of Internal Guide **Mrs. Prathibha P. M., Assistant Professor** and submitted in partial fulfillment of the requirements for the award of the Degree of Bachelor of Computer Applications by the **Dr. Manmohan Singh Bengaluru City University** during the academic year 2025-26. This report has not been submitted to any other Organization/University for any award of degree or certificate.

Name: NANDINI P.V.

Signature:

ACKNOWLEDGEMENT

I take this opportunity to first express my heartfelt gratitude to God for providing me with the strength, wisdom, and perseverance to successfully complete my internship and this report.

I would also like to sincerely thank my parents for their constant support, encouragement, and belief in me throughout this journey. Their guidance and motivation have been a strong foundation for my achievements.

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I would also like to express my deep appreciation to my internship guide **Mrs. Prathibha P. M.**, Assistant Professor, for their invaluable guidance, continuous feedback, and encouragement throughout the course of the internship. Their insights and support played a crucial role in successfully completing this experience.

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NANDINI P. V.

U18GH23S0113

ABSTRACT

The **Machine Learning-Based Disease Prediction with Mutation Awareness** project is an intelligent healthcare analytics system developed to predict diseases using healthcare and genomic datasets. The project integrates machine learning, mutation-aware analysis, anomaly detection, and healthcare visualization into a unified predictive healthcare platform capable of generating real-time disease prediction results and analytical insights.

The system is designed to support prediction for multiple diseases including breast cancer, heart disease, diabetes, and gene-based leukemia-related conditions such as ALL and AML. Healthcare and genomic datasets are collected and processed through data preprocessing techniques including missing value handling, normalization, transformation, and feature extraction. Feature engineering techniques are applied to convert raw healthcare and genomic data into structured analytical features suitable for machine learning prediction.

A major highlight of the project is the implementation of mutation-aware analysis, where genomic variations and abnormal gene expression patterns are analyzed to generate mutation-related features. These features improve the understanding of biological abnormalities associated with diseases and enhance the effectiveness of predictive healthcare analytics.

The system also incorporates anomaly detection techniques to identify unusual healthcare patterns and outlier behavior within datasets. Machine learning and hybrid prediction models are trained using processed healthcare data to generate disease prediction outputs with improved analytical performance. Multi-disease prediction functionality further allows simultaneous prediction of multiple diseases within a single platform.

The frontend dashboard provides an interactive environment for users to enter healthcare information, upload datasets, generate predictions, and analyze healthcare insights through graphical visualizations and analytical reports. The backend API manages communication between frontend interfaces and prediction models for efficient real-time processing.

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